Last Time

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \qquad P(B|A) = \frac{P(A \cap B)}{P(A)}$$

How to reverse of order of conditioning?

Bayes' Th.

$$P(A|B) = \frac{P(B|A)\,P(A)}{P(B)}$$

Here, we used the multiplication rule:

$$P(A \cap B) = P(A|B)\,P(B)$$
$$= P(B|A)\,P(A)$$

Independence

Definition: Two Independent Events

Events A and B are called **independent** if

$$P(A \cap B) = P(A)P(B).$$

when $P(A) \neq 0$ and $P(B) \neq 0$, we have

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B|A) = \frac{P(A \cap B)}{P(A)}$$

$$\therefore P(A \cap B) = P(A|B) \cdot P(B) = P(B|A) P(A)$$

If A and B are independent, we have

$$P(A \cap B) = P(A) P(B)$$

$$\therefore \quad P(A) P(B) = P(A|B) P(B)$$

$$P(A) P(B) = P(B|A) P(A)$$

$$\Rightarrow \quad P(A|B) = P(A)$$

$$P(B|A) = P(B)$$

$\Longrightarrow$ This states that knowing the event B occurs does not alter the probability of A. It is in this sense that the events are independent.

Example (Independent Events)
Suppose a fair coin is tossed twice. The sample space is

$$S = \{HH, HT, TH, TT\}.$$

Let

$$A = \{\text{the first toss is heads}\} = \{HH, HT\},$$

and

$$B = \{\text{the second toss is heads}\} = \{HH, TH\}.$$

***Question*:** Are the events A and B independent? Justify your answer using the definition of independence.

$A \cap B = \{HH\}$ $\therefore$ $P(A \cap B) = \frac{1}{4}$

$P(A) = \frac{|A|}{|S|} = \frac{2}{4} = \frac{1}{2}$

$P(B) = \frac{1}{2}$

$P(A \cap B) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2} = P(A)P(B)$

$\therefore$ A and B are independent.

Example (Independent Events with a Single Die)
Suppose a fair six-sided die is rolled once. The sample space is

$$S = \{1, 2, 3, 4, 5, 6\}.$$

Let

$$A = \{\text{the outcome is even}\} = \{2, 4, 6\},$$

and

$$B = \{\text{the outcome is divisible by } 3\} = \{3, 6\}.$$

***Question*:** Are the events A and B independent? Justify your answer using the definition of independence.

$A \cap B = \{6\}$

$P(A \cap B) = 1/6$

$P(A) = 3/6$

$P(B) = 2/6$

$P(A \cap B) = \frac{1}{6} = \frac{3}{6} \cdot \frac{2}{6} = P(A)P(B)$

$\therefore$ A and B are independent events.

COUNTING METHODS IN SEQUENTIAL EXPERIMENTS
-Determine the number of outcomes in the sample space

Example Consider the following **sequential experiment**, consisting of two subexperiments:

1. **Subexperiment 1:** Select one of two coins uniformly at random. Coin 1 is biased, with probability 3/4 of heads; Coin 2 is fair.
2. **Subexperiment 2:** Toss the selected coin once and observe either heads (h) or tails (t).

An outcome of the overall experiment records the outcomes of *both* subexperiments. Thus, the sample space is

$$\Omega = \{(1,h),(1,t),(2,h),(2,t)\}.$$

For example, $(1,h)$ means that Coin 1 is selected in the first subexperiment and a head is observed in the second.

Define the following events, which are subsets of Ω:

$$\begin{aligned} C_1 &= \{(1,h),(1,t)\}, && \text{Coin 1 is selected,} \\ C_2 &= \{(2,h),(2,t)\}, && \text{Coin 2 is selected,} \\ H &= \{(1,h),(2,h)\}, && \text{the toss results in a head,} \\ T &= \{(1,t),(2,t)\}, && \text{the toss results in a tail.} \end{aligned}$$

Since the coin is selected uniformly,

$$P(C_1) = P(C_2) = \frac{1}{2}.$$

The probabilities governing the second subexperiment depend on the outcome of the first:

$$P(H \mid C_1) = \frac{3}{4}, \qquad P(T \mid C_1) = \frac{1}{4},$$

whereas

$$P(H \mid C_2) = P(T \mid C_2) = \frac{1}{2}.$$

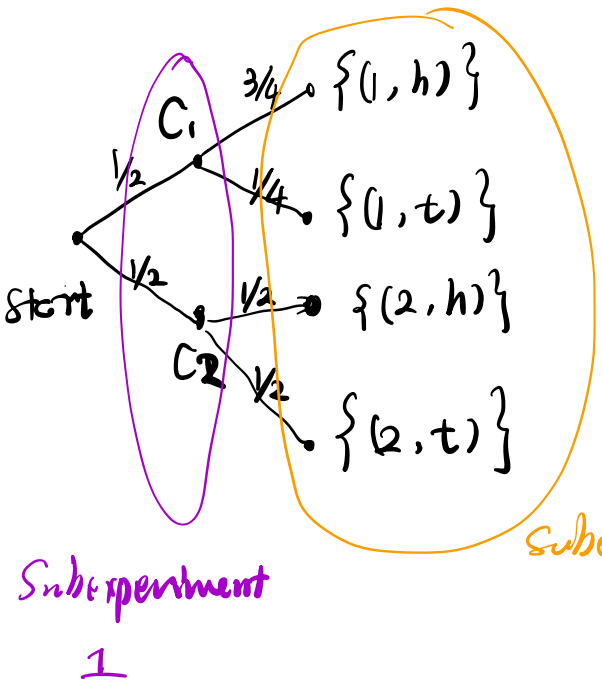
By the definition of conditional probability, we can get the probability of each outcome as an event:

$P(\{(1,h)\}) = P(C_1 \cap H) =$ $P(H \mid C_1)\,P(C_1) = \frac{3}{4} \cdot \frac{1}{2} = \frac{3}{8}$

$P(\{(1,t)\}) = P(C_1 \cap T) =$ $\frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$

$P(\{(2,h)\}) = P(C_2 \cap H) =$ $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$

$P(\{(2,t)\}) = P(C_2 \cap T) =$ $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$



Tree Diagram

$\{(1,h)\}$ and $\{(2,h)\}$ are mutually exclusive. (disjoint)

Then,

$$P\left(\underbrace{\{(1,h)\} \cup \{(2,h)\}}_{H}\right) \overset{\text{Axiom 3}}{=} P(\{(1,h)\}) + P(\{(2,h)\})$$

$$= 3/8 + 1/4$$

$H = \{(1,h), (2,h)\}$

Question 1. Given that the toss results in a head, what is the probability that Coin 1 was selected? That is, find

$$P(C_1 \mid H).$$

Defn. $\dfrac{P(C_1 \cap H)}{P(H)}$

$C_1 \cap H = \{(1,h)\}$

$P(C_1 \cap H) = \frac{3}{8}$

$= \dfrac{3/8}{3/8 + 1/4} = \dfrac{3}{5}$

Question 2. Given that the toss results in a tail, what is the probability that Coin 1 was selected? That is, find

$$P(C_1 \mid T).$$

Defn. $\dfrac{P(C_1 \cap T)}{P(T)}$

$= \dfrac{1/8}{1/8 + 1/4} = \dfrac{1}{3}$

A **sequential experiment** consists of two or more subexperiments performed in sequence. An outcome of the overall experiment records the outcome of each subexperiment.

In the above coin example,

$$\underbrace{\text{select a coin}}_{\text{Subexperiment 1}} \longrightarrow \underbrace{\text{toss the selected coin}}_{\text{Subexperiment 2}}.$$

Thus, for example,

$$(1,h) \in \Omega,$$

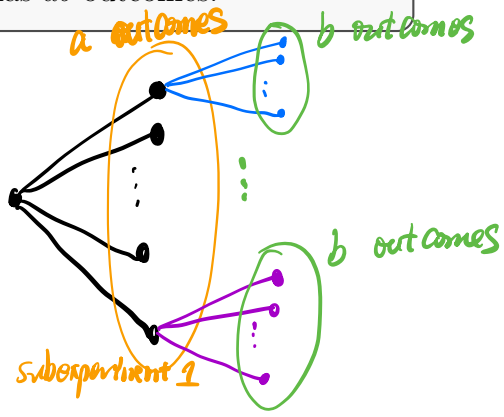
is an **outcome**, whereas

$$C_1 = \{(1,h),(1,t)\} \subset \Omega$$

is an **event**.

Generalization

An experiment consists of two subexperiments. If one subexperiment has a outcomes and the other subexperiment has b outcomes, then the experiment has ab outcomes.



Example We choose an ordered sequence of k distinguishable objects out of a set of n distinguishable objects. An ordered sequence of k distinguishable objects is called a **k-permutation**. What is the number of k-permutations of these n distinguishable objects?